

Problem 6

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Problem 1 A piecewise defined function f is given by

$$f(x) = \begin{cases} \frac{2x-3}{x-2} & \text{if } x < 2 \\ \frac{x^2-5x+6}{x^2-4} & \text{if } x > 2 \end{cases}$$

Determine the form of the limit, then find the limit.

- (a) $\lim_{x \rightarrow 2^+} f(x)$
- (b) $\lim_{x \rightarrow 2^-} f(x)$
- (c) $\lim_{x \rightarrow 2} f(x)$